

What OECD well-being data from before and during the pandemic can tell governments about responding to crisis

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Abstract

Governments increasingly use well-being indicators to steer policy [2], but sudden and unforeseen events can raise a practical question: “Do the factors that track population well-being in normal times still hold when circumstances change abruptly?” We use the OECD How’s Life? database to compare two methods for choosing the most relevant measures. The first was a data-driven approach which relied on a purely statistical method, namely the strongest cross-country correlates of life satisfaction in the pre-pandemic years 2013 and 2018, taking one indicator per well-being domain. The other approach was a theory-driven set of five indicators selected from Maslow’s hierarchy of needs [7]. We first stage a head-to-head on life satisfaction. We then extend the data-driven method to two further outcomes namely negative affect (emotional) [5] and deaths from suicide, alcohol and drugs (behavioural). All models are evaluated in the pandemic years 2021–2022, which the screen never saw. Our findings indicate the data-driven screen beats the theory on every outcome, decisively so for life satisfaction (adjusted R^2 0.85 vs 0.29). While each of the three outcomes relies on a different aspect of well-being, social connections for overall life satisfaction, economic activity for negative emotions, and education or skill levels for deaths of despair, no single group of measures covers them all. Fundamentally, what matters to people does not change during a crisis. Statistical tests show that the relationship between these factors and well-being remained identical before and during the pandemic.

Keywords

Well-being; OECD How’s Life?; life satisfaction; COVID-19; Maslow’s hierarchy of needs;

Acronyms

OECD: Organisation for Economic Co-operation and Development. Provides the “How’s Life?” database. **SDMX:** Statistical Data and Metadata eXchange. This is the ISO standard the OECD uses to publish data and defines the OBS_STATUS quality flags. **CWB / FWB:** Current Well-being / Future Well-being (The two OECD dataflows files this study draws on). **OLS:** Ordinary Least Squares. The linear regression method used for all models. **CI:** Confidence Interval. The range within which the true value plausibly lies (95% throughout). **VIF:** Variance Inflation Factor. Measures how much a predictor overlaps with the other predictors. High values make individual coefficients unreliable. **SD:** Standard Deviation. Coefficients are reported per 1 SD change in a predictor. **EU-SILC:** European Union Statistics on Income and Living Conditions. **DFBETA:** “Difference in beta” measures how much one coefficient changes if a single country is removed, in standard-error units. **NPISH:** Non-Profit Institutions Serving Households. Appears in the income indicator name. **WISE:** The OECD Centre on Well-being, Inclusion, Sustainability and Equal Opportunity.

Research question

“Given what national statistics say about people’s perceived well-being and whether or not these drivers differ in times of crisis, how can governments best respond to the needs of their people?”

Operationally, we decompose this into three answerable sub-questions:

1. **Which observable indicators best track the population’s wellbeing in normal times?** We answer this by following two different approaches:
 - **data-driven:** a correlation analysis over all viable OECD wellbeing indicators
 - **theory-driven:** five indicators chosen a priori to represent the levels of Maslow’s hierarchy of needs [7]
2. **Do those relationships survive a systemic crisis?** The COVID-19 pandemic provides a natural before/after design. The years 2013 and 2018 are considered as ordinary, while 2021 and 2022 are crisis years.
3. **Is one general measurement framework enough, or does each dimension of well-being need its own tailored set of measures?** Comparing a single set of theory-based measures against custom, data-driven sets shows us where a one-size-fits-all approach works, where it falls short, and which key measures are backed by both methods.

Definitions

Life satisfaction: Life satisfaction is measured through survey questions concerning overall

satisfaction with life. The OECD measures this subjective indicator by asking questions like, “Overall, how satisfied are you with your life as a whole these days?”. Mean values on an 11-point scale, with responses ranging from 0 (not at all satisfied) to 10 (fully satisfied) are recorded with slight variations of the exact question asked based on the country performing the survey.

Negative affect balance: This measures the percentage of people in a country’s population reporting more negative than positive emotions the previous day. The emotions typically asked about in surveys centre around feelings of anger, sadness, worry, how much a person smiled, laughed or how well rested an individual feels. Data samples one thousand individuals per country from the Gallup World Poll [4].

Deaths of despair [8]: These are register-based deaths from suicide, alcohol abuse and drug use per 100 000 (modelled in logs) age-standardised based on the 2015 OECD population structure. This is a behavioural outcome independent of survey self-report and covers death by poisoning or exposure to noxious substances.

Screen window: The screen window in our study refers to the country means calculated for 2013 and 2018. These years are considered “normal” years compared to the fit window due to the lack of major, global incidents at the scale seen between 2020 and 2022 during the height of the COVID pandemic.

Fit window: Refers to country means calculated between 2021 and 2022. This is the anomalous window where we aim to compare the screen window against to see if there exists a measurable difference in life satisfaction, negative affect or deaths from suicide, alcohol and drugs.

Driver: this is shorthand for any cross-sectional statistical association. These are not used to

make any causal claims in our conclusions but point to relationships that may exist in the years those indicators are recorded.

Motivation and stakeholders

The primary stakeholder is government policy units practising well-being-oriented budgeting. An example could be New Zealand's Wellbeing Budget, the OECD WISE Centre's policy work [2]. In times of crisis, various government entities need to make decisions on where to divert their resources. This study aims to understand if there is empirical evidence on where best to divert resources and if there are significant indicators to suggest a different approach from the one it already applies. The models we developed are meant to answer these questions. If pre-crisis drivers still predict crisis-era well-being with stable coefficients, then the response to a crisis is the same normal playbook, just executed under scenarios of extreme pressure.

The data

The basis of this research is made up from the data contained in two files downloaded from the OECD Well-being Data Portal [1]:

- **Current Well-being (CWB):** 111266 rows, 68 measures across 11 domains (Income & wealth, Work & job quality, Housing, Work-life balance, Health, Knowledge & skills, Social connections, Civic engagement, Environmental quality, Safety, Subjective well-being), 47 countries, years from 2004 to 2026. CWB carries the well-being outcomes and lived-experience predictors.
- **Future Well-being (FWB):** 37858 rows, 35 measures across 4 capital domains (Economic, Natural, Human, Social capital), 48 countries.

Every indicator's observed value carries SDMX OBS_STATUS quality flags [3]. The distribution of the data according to SDMX classifications is such that:

- ~90% are A (normal)
- ~7% are D (definition differs footnote-level deviation)
- ~1% are B (time-series break)
- ~2% of retained observations carry B or D flags.

Survey based measures which include life satisfaction, negative affect, trust, satisfaction items, originate largely from the Gallup World Poll and EU-SILC waves. Mortality, income and labour measures are register/administrative indicators obtained from static records. Life satisfaction is only collected in certain time periods and is available for all countries only in the years 2013, 2018, 2021 and 2022, which determines our two windows.

Strengths of the Data

The dataset aligns well with policy decision making because it draws from official, internationally harmonized indicators across 11 life domains and four capitals. Using an independent, register based outcome eliminates common source bias from survey data. Additionally, the temporal span covers periods before and after the pandemic, while integrated SDMX quality flags ensure any data issues are explicitly highlighted rather than hidden.

Weaknesses of the Data

(i) The unit of analysis is made up from 33 countries that have both survey outcomes in both windows, capping model complexity at 5 predictors. This is workable for developing our models but more data points would reduce outliers.

(ii) Life satisfaction, negative affect and several strong predictors share the Gallup source and part of their intercorrelation is common-method variance. How the data is collected links certain indicators together by their method of collection.

(iii) The samples collected are predominantly European. Findings generalise to developed, mostly-European OECD economies due to a lack of data from other OECD countries.

(iv) Sex and education disaggregated rows coexist with totals in the raw export. Simple deduplication silently mixes populations (we found e.g. Argentina 2016 employment rate: 59.9% female, 86.6% male, 72.4% total). We filter all three dimensions to Total explicitly.

(v) Correlations are cross-sectional at country level with no causal identification, and no individual-level inference (ecological fallacy).

Methodology

We split the data in time, screening (2013–2018) and model evaluation (2021–2022) addressing two core issues:

- **Methodological Integrity:** It prevents post-selection bias ensuring reported R^2 values are not artificially inflated by selecting predictors on the same dataset used to evaluate them.
- **Structural Stability:** Because the cutoff coincides with the COVID-19 pandemic, cross-window validation tests whether pre-pandemic relationship drivers remain stable during a crisis or if their underlying coefficients shift.

One consequence is that for every data-driven model, the screen-window fit is in-sample by construction. The indicators were chosen on that window against that outcome, and so screen

window R^2 values are optimistic benchmarks. The honest performance number is always the fit-window one, and the informative quantity is the drop between the two.

Data Cleaning

We filter AGE/SEX/EDUCATION to Total jointly, dropped measures reported in multiple units across age ranges and male/female/total populations. OBS_STATUS we audited to ensure we aren't introducing bad data by mixing data of different definitions and time series breaks. We also pivot our data to one row per country-year, one column per measure. We also collapse each window to country means.

Coverage filters

The core sample was made up of 33 countries with both survey outcomes in both the 2013/18 and 2021/22 windows. This identical sampling on both sides of the comparison ensures our results are not affected by our sample composition. The candidate pool was made of all measures outside the subjective well-being domain. The outcome we are modeling for must not be part of the predictors. Just like life satisfaction, deaths of despair are likewise excluded from the pool because it is an outcome. Indicators were deemed as viable if they had greater than an 80% coverage of the core sample in both windows.

Data-driven screening Models

For each of the three outcomes separately, we select the top five predictors for these models. We calculated the correlation between each candidate indicator and each individual well-being outcome (life satisfaction, negative affect balance, deaths of despair) over the 2013–2018 period (using 95% confidence intervals and sample sizes for each pair).

Instead of simply picking the five highest numbers overall, we filtered them by broad category to make sure we weren't choosing multiple indicators that essentially measure the exact same thing. We grouped the candidates by domain, and picked the single strongest performer from each domain, and then selected the top five indicators from that narrowed list.

We capped the final selection at five to match the five factors used in the Maslow framework. This limit was decided during the initial design phase, before we had run any comparisons, to ensure the cutoff wasn't adjusted after looking at the results.

Theory-driven Maslow Model

Maslow's theory of needs [7] is a relatively old theory, but it remains highly influential and continues to provide a conceptual framework for many approaches to human development and public policy [11].

The theory identifies five categories of human needs namely physiological needs, the need to feel safe, the need for love and belonging, esteem among one's peers and social group, and self-actualization. These needs are organized hierarchically, from the most basic to the higher order needs.

Unlike the data-driven arm, this single set is applied unchanged to all three outcomes. This makes Maslow our general-purpose theoretical framework.

Maslow's Theory and Indicator Selection

The five indicators are mapped a priori to the levels of Maslow's hierarchy and justified theoretically, rather than by their correlations to life satisfaction. This matches the theoretical approach rather than fitting any given data point to the correlation outcomes.

Life expectancy at birth meets the physiological and basic bodily needs met at population scale. The physiological need for human survival, particularly health and physical well-being, make this indicator a relevant proxy and reflect the general mortality conditions of a population and through children born.

Housing affordability is our proxy for a person's need for stability and secure living conditions. Having a place to stay is a fundamental component of a person's security. Excessive housing costs can create financial and residential instability and pose a direct impediment to that safety. The OECD indicator measures the percentage of household disposable income that remains after housing costs, including rent and maintenance. A higher proportion of income remaining after these costs can therefore indicate greater financial security.

Social support covers the need to belong, interpersonal relationships and the social network. The OECD measures social support through the percentage of people who report having friends or relatives they can rely on in times of trouble.

A country's employment rate can be taken as a measure of the percentage of people who seek the validation and esteem of others through gainful work. Employment can contribute to esteem by providing individuals with a sense of achievement, independence, social recognition, and personal value.

Self actualization represents the highest level of Maslow's hierarchy and concerns personal fulfillment and the realization of one's potential. Self actualisation can be measured through the rates of volunteering through organisations. Volunteering as an act not aimed at self improvement and not motivated by monetary reward shows one's ability to act beyond their basic needs. One tends not to volunteer if their

main priority is meeting needs of security.

The Modeling Process

For each indicator set we estimate three separate OLS regressions with one per outcome (life satisfaction, negative affect, log deaths of despair). Because the specification never changes, differences in performance across outcomes are attributable to the outcomes themselves, and differences between the two arms (Maslow versus Data Driven) are attributable to the indicator choice alone.

Before fitting, each predictor is z-standardised (rescaled to mean 0, standard deviation 1). The predictors arrive in incompatible units, namely percentages, years of life expectancy and 0–10 survey scores, so raw coefficients would not be comparable.

After standardisation every coefficient answers the same question, “how much does the outcome differ between two countries one standard deviation apart on this indicator, holding the other four constant?” Outcomes are kept in their native units, so coefficient magnitudes are not comparable across outcomes. Life satisfaction is on a 0–10 scale while the negative affect is a percentage share. A β of -2.6 on the affect model means 2.6 percentage points, which is not a huge effect.

Deaths of despair are modelled in logs because it is a skewed rate. The log form treats a doubling of deaths the same everywhere, rather than letting a few high-mortality countries dominate.

Each model is estimated on complete cases with countries with values for the outcome and all five predictors in the fit window. This is why the number of countries (n) varies slightly by model (23–33). For each model we report:

- **Adjusted R^2 :** the share of cross-country variance explained, penalised for the number of predictors. The penalty matters at $n \approx 23$ –33, where unadjusted R^2 rises mechanically with every added variable.
- **Coefficient confidence intervals:** the range of effect sizes compatible with the data, more informative than a bare significance star at this sample size.
- **VIFs:** Which checks for predictor overlap. When predictors are highly intercorrelated, the regression cannot cleanly attribute shared variance to one of them, and individual coefficients become unstable even while the model's overall fit remains trustworthy. Where values are above ~ 5 we rest claims on blocks of predictors rather than single coefficients.
- **A nested-block ΔR^2 F-test:** we first fit the material-conditions block alone, then add the social/relational block and formally test whether the improvement in fit is larger than chance. This answers the policy-relevant question directly. “Do social factors explain well-being beyond material conditions?”
- **A screen-window benchmark:** the identical specification refit on 2013–18 means. For the screened arm this benchmark is optimistic by construction (the indicators were chosen on that window), so the informative quantity is the drop from benchmark to fit window. A small drop means the relationships transferred well to the pandemic years.
- **Pooled coefficient stability test (Chow style test):** the formal answer to “did the drivers change?”. We stack both

windows into one dataset, add a pandemic-era dummy variable and its interaction with every predictor, and jointly F-test the interactions. If the pre-pandemic coefficients still describe the pandemic era, the interactions are indistinguishable from zero. One caveat, each country contributes an observation to both windows, so these observations are not fully independent and the p-values are heuristic rather than exact.

Residual analysis

A residual is the gap between what a country reports and what the model predicts from its conditions (observed minus predicted). A positive residual means a country is more satisfied than its indicators suggest while a negative one, shows less satisfied than the indicators predict. Residuals are where the model admits what it does not know, which makes them the most policy-relevant part of the output after the coefficients themselves. A large residual flags a country whose well-being is being driven by something outside the dashboard.

We use them in two steps. First, we rank countries by residuals and flag those misfit by more than one standard deviation of the residuals. These are the over and under performers reported in Results.

Comparison Design

The comparison holds everything constant except indicator choice. Maslow uses the same 33-country core sample, the same screen and fit windows collapsed to country means, the same three outcomes, the same OLS form with z-standardised predictors, and the same reporting. Under those conditions, a performance gap between the arms is evidence about selection strategy (data-driven screening

Second, we correlate the residuals with the viable indicators held out of the model. A strong correlation identifies a candidate omitted variable. (A factor that systematically lines up with the model's misses). With $n \approx 25\text{--}30$ per correlation we treat $|r| > 0.35$ as "worth discussing", not as proof.

Because a large residual can mean very different things, we follow up with standard influence diagnostics, each answering a distinct question. Studentized residuals with a Bonferroni correction ask, "is this miss statistically surprising, given that in any sample of 27 someone must be furthest from the line?"

Leverage asks, "is the country unusual in its conditions (predictor values), such that it pulls the fitted line toward itself?" Cook's distance and DFBETAs ask, "if we deleted this country, would the model or any single coefficient, change materially?"

Where the diagnostics implicate a specific omitted factor, an added-variable check (refitting with that factor included) tests whether it actually absorbs the miss. Where they implicate a single influential country, a leave-one-out refit shows what the model looks like without it. The Outlier Analysis in Results applies exactly this sequence to Latvia and Greece and shows they are opposite cases. A genuine response outlier versus a high-leverage point

versus theory-driven mapping) rather than incidental setup differences.

Three asymmetries must be kept in view when reading the comparison table:

- **Maslow and the Data Driven model differ by design.** The data-driven arm fields a different five-indicator set per outcome. The Maslow arm fields one fixed set. This is to address sub-question

3 where one theory that claims to describe human needs in general should not require re-selection per outcome, and the comparison measures what that generality costs.

- **Screen-window fits are in-sample for the data-driven arm.** Every screened set's 2013–18 R^2 is flattered by construction; the Maslow arm's is not (its indicators were chosen by theory). The fairest single point of comparison is the fit window, where neither arm has seen the data.
- **Complete-case counts differ slightly** (Maslow $n = 30\text{--}33$ vs screened $n = 23\text{--}28$). Because the Maslow indicators happen to have fuller coverage, there is a small, disclosed precision advantage for the theory arm.

Within Maslow's theory, life satisfaction is the outcome the hierarchy culminates in self-actualization and ties directly to life satisfaction, so the life-satisfaction contest is the theory's home ground and the head-to-head we headline. The extensions to negative affect and deaths of despair then ask how far each approach travels beyond it.

Finally, we report the overlap between the

indicator sets. An indicator selected by both the screen and the theory is empirically and theoretically warranted. The employment rate appears both in the Maslow set (esteem) and in the negative-affect screen and the sets otherwise split along measurement lines. The life-satisfaction screen is entirely survey-perception based, the Maslow set largely register and administrative. That split is what lets the life-satisfaction comparison double as a check on common-source inflation.

Tools

- Python and pandas used for reshaping and window means
- statsmodels for OLS, F-tests and VIFs
- matplotlib for figures
- Jupyter notebooks for running the full pipeline end-to-end

We deliberately use interpretable linear models rather than Machine Learning algorithms. With 27 complete cases, flexible learners would overfit and cross-validation would be unstable. The stakeholder question requires coefficients with confidence intervals, not black-box predictions. The methodological effort went into the design, selection of indicator and evaluation split, same sample windows and stability testing rather than into model competition

Results

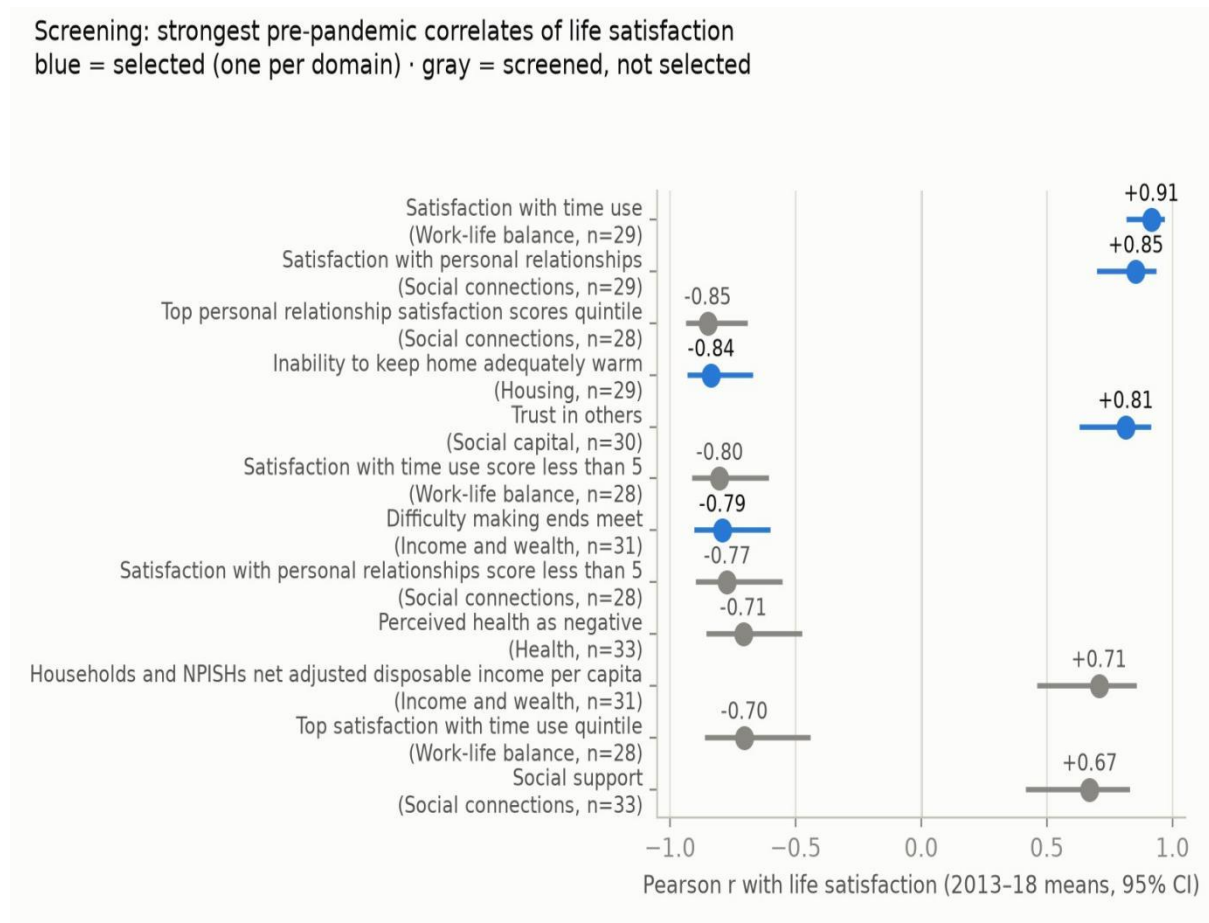


Figure 1. The pre-pandemic screen for life satisfaction. The strongest cross-country correlates between 2013–18 means with 95% confidence intervals. Filled markers are the five selected (top |r| per domain). The rest are the strongest runners-up.

Model 1: Life Satisfaction

The model evaluates cognitive life satisfaction

during the 2021–2022 pandemic fit window (sample size $N = 27$). The model demonstrates exceptional explanatory power, achieving an R^2 of 0.8785 and an Adjusted R^2 of 0.8496. This performance closely mirrors the pre-pandemic screen window benchmark (2013–2018 Adjusted $R^2 = 0.87$), showing that the selected indicators' predictive strength carries forward.

Selected Predictor	Standardized β (1SD)	p-value	95% CI Lower	95% CI Upper
Satisfaction with time use	0.148	0.212	-0.091	0.387
Satisfaction with personal relationships	0.293	0.004	0.105	0.481
Inability to keep home adequately warm	-0.101	0.174	-0.251	0.048
Trust in others	0.153	0.023	0.024	0.282
Difficulty making ends meet	0.048	0.526	-0.108	0.205

Cognitive life satisfaction during the pandemic was overwhelmingly driven by social capital and relational stability. 'Satisfaction with personal relationships' emerged as the single strongest and most statistically significant predictor ($\beta = 0.293$, $p = 0.004$, 95% CI [0.105, 0.481]), followed by 'Trust in others' ($\beta = 0.153$, $p = 0.023$, 95% CI [0.024, 0.282]). These findings highlight that during acute external shocks, cognitive evaluation of life is heavily anchored in social support networks and on institutional and interpersonal trust, whereas material deprivation variables ('Inability to keep home warm' and 'Difficulty making ends meet') played secondary, statistically non-significant direct roles once relational buffers were accounted for.

The nested-block test shows the material block alone reaches adjusted R^2 0.53, and adding the social/relational block lifts it to 0.85 ($F = 17.8$, $p < 0.001$). Social factors add large, formally

significant explanatory power beyond material conditions.

Model 2: Negative affect balance

Model 2 examines negative affect balance during 2021–2022 (sample size $N = 28$). The model yields an R^2 of 0.6978 and an Adjusted R^2 of 0.6292. In comparison to the pre-pandemic benchmark (2013–2018 Adjusted $R^2 = 0.86$). This model shows the largest benchmark-to-fit drop in the study ($0.86 \rightarrow 0.63$). Two mechanisms plausibly contribute and the design cannot separate them. A genuine disruption of day-to-day emotional experience during the pandemic, and selection-bias decay, since the benchmark is in-sample for this screen too. What can be said precisely is that the drop is far larger than the life-satisfaction model's (0.02) on an identical design.

Selected Predictor	Standardized β (1SD)	p-value	95% CI Lower	95% CI Upper
Corruption	-1.744	0.134	-4.069	0.581
Satisfaction with time use	-2.544	0.051	-5.096	0.008
Difficulty making ends meet	-1.622	0.206	-4.205	0.961
Households with internet access at home	-0.320	0.748	-2.362	1.722
Employment rate	-2.557	0.020	-4.663	-0.450

Hedonic emotional states are strongly tied to daily operational stability and economic participation. The 'Employment rate' is the primary statistically significant anchor ($\beta = -2.557$, $p = 0.020$, 95% CI [-4.663, -0.450]), indicating that higher national employment significantly mitigated negative affect/distress. 'Satisfaction with time use' approached statistical significance ($\beta = -2.544$, $p = 0.051$, 95% CI [-5.096, 0.008]).

Model 3: log Deaths from Despair

Model 3 models mortality from deaths of despair (suicide, alcohol, drug overdoses) on a log scale (sample size $N = 23$). The model produces an R^2 of 0.5377 and an Adjusted R^2 of 0.4017 during 2021–2022, compared to an Adjusted R^2 benchmark of 0.55 in the 2013–2018 period.

Selected Predictor	Standardized β (1SD)	p-value	95% CI Lower	95% CI Upper
Educational attainment among young adults	0.133	0.173	-0.064	0.330
Full-time employees earning < 2/3 gross median	0.083	0.391	-0.116	0.281
Student science skills	0.189	0.052	-0.002	0.380
Perceived health as positive	-0.131	0.198	-0.338	0.075
Disposable income per capita (NPISHs net)	-0.020	0.849	-0.238	0.198

Objective behavioral pathology metrics like deaths of despair exhibit greater structural noise during public health emergencies. 'Student science skills' (a proxy for broader educational infrastructure and human capital) was the strongest predictor, approaching significance ($\beta = 0.189$, $p = 0.052$, 95% CI [-0.002, 0.380]). Short-term disposable income showed negligible direct explanatory power ($\beta = -0.020$, $p = 0.849$), demonstrating that severe behavioral distress is driven by long-term structural and educational

determinants rather than immediate macroeconomic transfers [8].

The model explains a moderate 40% of cross-country variance, far from the life-satisfaction model's 85%. It is still a substantive improvement over what any general well-being set achieves on this outcome (the Maslow set reaches 0.21). This model carries the study's weakest data position ($n = 23$, below the five-observations-per-predictor guideline)

Maslow Comparison against Data-Driven Models

Outcome	Data-Driven R2	Data-Driven Adj R2	Data-Driven N	Maslow R2	Maslow Adj R2	Maslow N
Life satisfaction (cognitive)	0.8785	0.8496	27	0.3995	0.2883	33
Negative affect balance (hedonic)	0.6978	0.6292	28	0.6016	0.5278	33
log Deaths of despair (behavioural)	0.5377	0.4017	23	0.3475	0.2116	30

The data-driven arm wins on every outcome, but the margin varies enormously. 0.56 on life satisfaction, 0.10 on negative affect, 0.19 on deaths of despair. The pattern is informative. Where outcome and predictors share a survey source (life satisfaction), the gap is huge. Where they do not, it shrinks to modest. The Maslow set is genuinely competitive on negative affect, driven largely by the one indicator both arms selected, the employment rate. Maslow is also weakest on the mortality outcome, where its best predictor is the one closest to the outcome's construct (life expectancy).

The Maslow set on life satisfaction (n = 33) reaches adjusted $R^2 = 0.29$. The theory arm's five indicators capture about a third of the cross-country variance the screened set captures. The screened set's advantage of 0.56 in adjusted R^2 is the largest gap anywhere in the study, and it occurs on the outcome the hierarchy was built to explain.

Part of this performance gap reflects a genuine

difference, but part comes with an important limitation. Relational perceptions (relationships, trust) genuinely outperform register-based proxies (life expectancy, housing affordability) in tracking evaluative well-being across countries. Life satisfaction and several screened predictors share the Gallup survey source, so part of the advantage is common-source inflation. The negative-affect and deaths-of-despair models above, where the outcomes are *not* measured by the same instruments as the screened predictors, bound how much of the story that caveat can carry.

One Screened Set Per Outcome Shown Visually

Life satisfaction recruits relational and perceived-adequacy indicators. Negative affect recruits economic-participation and institutional indicators. Deaths of despair recruits human-capital and health indicators. No indicator appears in all three screens.

Which indicators each model uses

blue = outcome-specific data-driven screens · orange = Maslow set
ringed = selected by more than one model

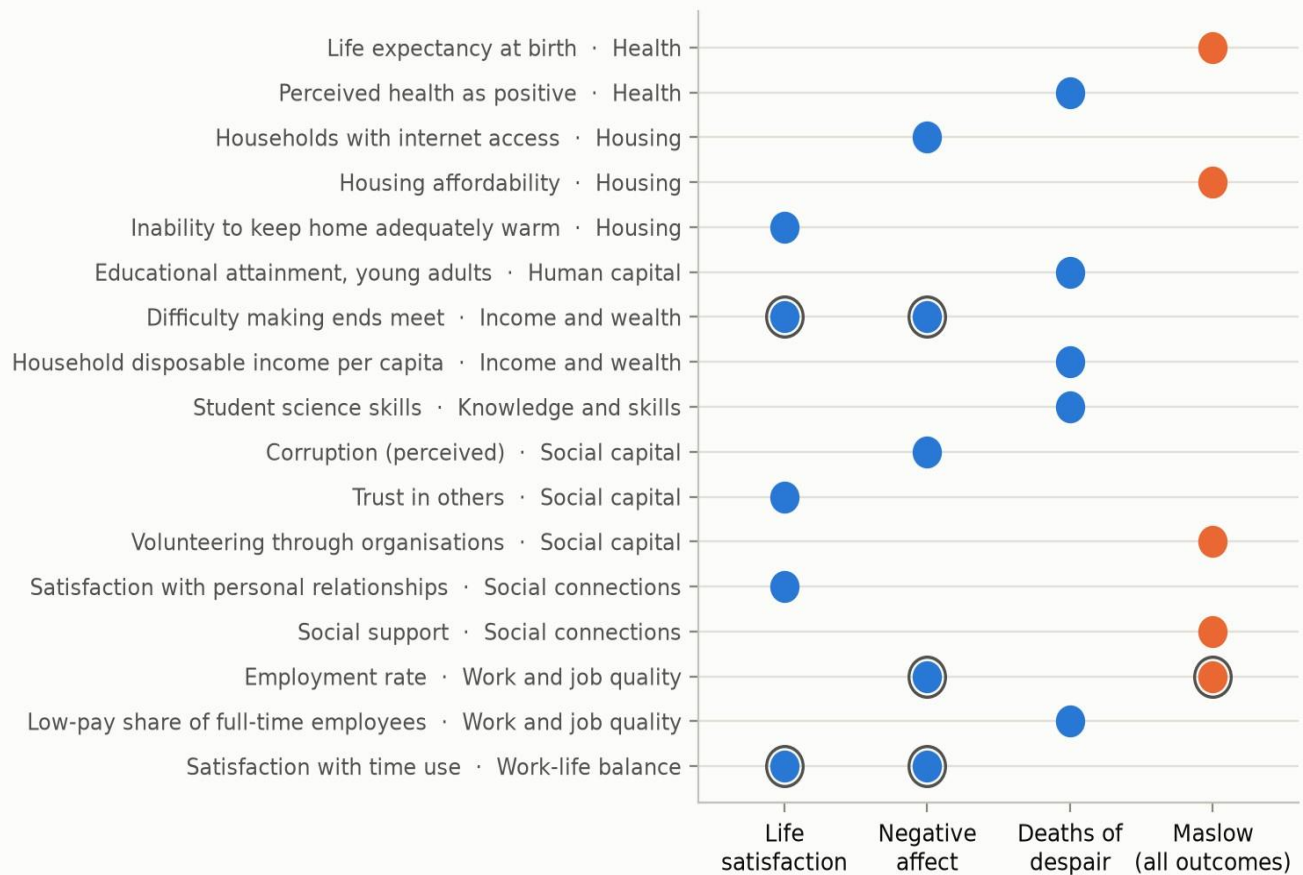


Figure 2. Which indicators each model uses, grouped by the well-being domain. Blue: the three outcome-specific data-driven screens; orange: the fixed Maslow set. Ringed marks are indicators selected by more than one model. The employment rate is the only indicator shared between the theory arm and a screen.

Stability: nothing re-ranked

The pooled Chow test finds no detectable

coefficient shift in any of the six models: interaction p-values are 0.42, 0.56 and 0.86 for the screened life-satisfaction, negative-affect and deaths-of-despair models, and 0.69, 0.80 and 0.86 for the corresponding Maslow models. Within the resolution of these tests (and their disclosed non-independence caveat), the pre-pandemic structure of what tracks well-being carried into the pandemic unchanged, in both arms. Levels moved; the map did not.

Coefficient stability on life satisfaction across the two windows
joint predictor $\times$ window interaction test: $p = 0.41$

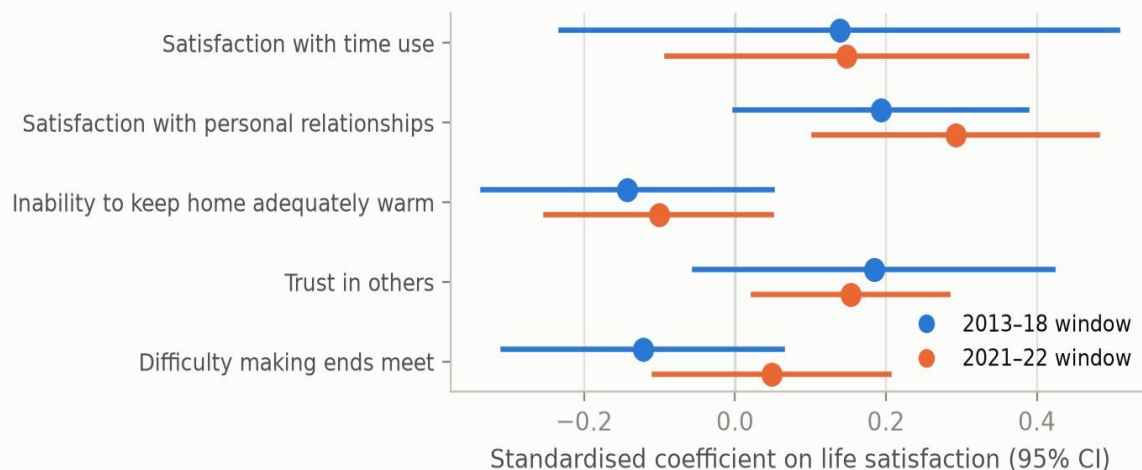


Figure 3. Coefficient stability for the life-satisfaction model: standardized coefficients with 95% confidence intervals, estimated

separately on the screen window and the fit window. Every pair overlaps; the pooled interaction test confirms no detectable shift.

Residuals (fit window / life satisfaction)

headline deviations are different phenomena.

Over-performers:

- Greece (+0.29)
- Luxembourg (+0.28)
- Italy (+0.23)
- Austria (+0.20)
- Romania (+0.19)

Under-performers:

- Latvia (-0.66)
- Portugal (-0.27)
- Hungary (-0.26)
- Ireland (-0.26)

Residuals correlate most with perceived health ($|r| \approx 0.45$) and gross fixed capital formation (+0.48).

Outlier Analysis

Standard influence diagnostics (studentized residuals with Bonferroni outlier test, leverage, Cook's distance, DFBETAs) show the two

Latvia is a genuine response outlier

The only significant one in the sample (studentized residual -3.8 , Bonferroni $p = 0.03$). Its leverage is low (0.10). Its measured conditions are ordinary, even good (above-average trust in others and time-use satisfaction), yet reported satisfaction is far below prediction. An added-variable check on the factor the residuals implicate perceived health, where Latvia is the sample's worst at 50% positive vs. a 68% mean, absorbs roughly 40% of the miss (residual $-0.66 \rightarrow -0.38$, health $\beta = +0.13$ per SD, $p = 0.015$; model adj. R^2 0.85 $\rightarrow$ 0.88). The remainder is consistent with the post-transition life satisfaction gap documented for ex-Soviet economies [10]

Greece is not a significant outlier:

(Bonferroni $p = 0.22$) but the sample's most influential point, leverage 0.76 and Cook's $D = 3.3$, driven by economic strain nearly four

standard deviations above the sample mean (71% report difficulty making ends meet). One DFBETA is extreme, dropping Greece moves the ends-meet coefficient by 4.1 standard errors, from a puzzling +0.05 ($p = 0.53$) to a theory-consistent -0.14 ($p = 0.06$). Greece's apparent "over-performance" is thus a high-leverage extrapolation effect, a caution

about the model's linearity at extreme deprivation (and consistent with adaptation to a prolonged crisis) rather than evidence that Greeks are anomalously happy. The wrong-signed ends-meet coefficient in the headline model is a Greece artefact and is reported as such

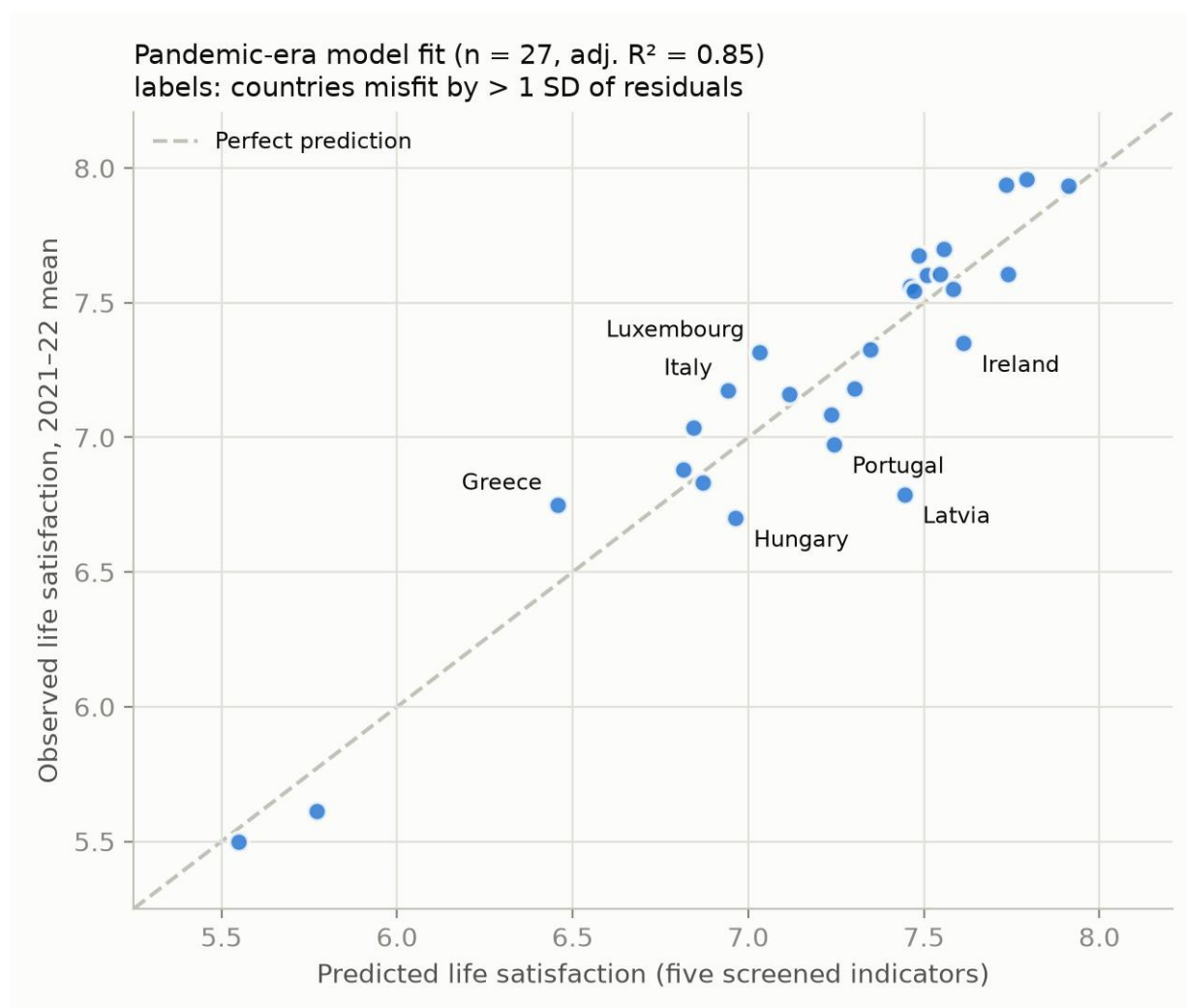


Figure 4. Pandemic-era model fit for life satisfaction ($n = 27$, adjusted $R^2 = 0.85$). Countries above the dashed identity line report higher life satisfaction than their conditions predict. Labels mark countries misfit by more than one standard deviation of the residuals. Latvia is the largest outlier

Discussion

1. The drivers held:

A predictor set frozen on 2013/2018 data explains pandemic-era life satisfaction essentially as well as it explained

pre-pandemic life satisfaction, with no detectable coefficient shift. The pandemic changed **levels** of well-being in particular countries (the residual analysis identifies which), but not the cross-country **structure** of what tracks it. For the stakeholder question, the

implication is direct. Crisis response does not require a new dashboard, it requires protecting the same fundamentals harder.

2. Relational factors are not a luxury:

Social/relational indicators add large explanatory power beyond material conditions for the cognitive outcome and significant power for the behavioural one. This is consistent with the World Happiness Report literature [6]. The perceived adequacy form of material indicators (can you keep your home warm, can you make ends meet) outperformed income levels in the data driven screen model [9]. The policy translation suggests monetary transfers and energy support that restore adequacy, and infrastructure that sustains social connection, are first-order well-being policy, not secondary priorities.

3. Each dimension of well-being has its own set of indicators:

The three screens select three almost disjoint indicator sets where relational perceptions for evaluative well-being, economic participation for hedonic well-being, human capital and health for behavioural harm. No indicator tracks all three outcomes, and a set tuned to one outcome does poorly on the others. In particular, deaths of despair are barely tracked by general well-being indicators (Maslow adj. R^2 0.21) and become moderately trackable (0.40) once screened on their own terms, through educational and labour-market structure. Governments should therefore not assume that policies protecting evaluative well-being automatically protect against behavioural harm. The

latter needs its own indicators and dedicated public-health surveillance [5], and this study identifies concrete candidates for that watch-list.

4. Theory vs. Data:

The screened (perception-based) set beats the Maslow (largely register-based) set on every outcome, but the margins tell the story. 0.56 of adjusted R^2 on life satisfaction, 0.10 on negative affect, 0.19 on deaths of despair.

Two readings coexist and the design cannot fully separate them. Outcome-tailored perception indicators genuinely predict better than one fixed register-based set, and part of the life-satisfaction gap specifically is common-source inflation (that outcome shares the Gallup instrument with its predictors, the other two do not). Crucially, the screen and the hierarchy converge in the middle of the pyramid. Belonging-type indicators (relationships, trust, social support) carry weight in both arms, and the single indicator selected by both approaches (employment) anchors the hedonic outcome in both. The theory's lower levels (life expectancy, housing) under-perform their billing in cross-country data, with each Maslow indicator doing best on the outcome nearest its construct (employment $\rightarrow$ daily affect, life expectancy $\rightarrow$ mortality). Both arms pass the stability test, so the theory-vs-data ranking is not a pandemic artefact.

5. Methodological Robustness and Limitations

The reliability of these findings rests on a rigorous evaluation design. By enforcing

a strict pre- and post-pandemic split on an identical country sample, and integrating a register-based outcome alongside survey data, we minimized the risk of post-selection bias. We also prioritized simple, interpretable models over complex black-box algorithms, ensuring the results remain transparent for policymakers. Nevertheless, certain limitations must be acknowledged. Our analysis is constrained by a small, predominantly European sample size (around 30 countries), which limits the global generalizability of our findings. Additionally, the reliance on a common survey source for both life satisfaction and several predictors inevitably introduces a degree of common-method variance. The presence of predictor intercorrelation also means that while our models are highly predictive overall, individual coefficients should be interpreted with caution. Finally, calculating means across the 2021–2022 window blurs some temporal nuances, and our pooled stability tests treat overlapping country data as independent, which warrants a conservative reading of the p-values.

Conclusion

This study provides a clear directive for policymakers. We see that a systemic crisis does not require a wholesale redesign of well-being monitoring. Our findings demonstrate that the fundamental drivers of well-being remained stable throughout the pandemic. The core "map" of what determines well-being, perceived material adequacy and the relational fabric of society, did not re-rank under pressure. Rather than searching for new, crisis-specific metrics, governments should focus on protecting these established fundamentals with greater

intensity.

Furthermore, our findings demonstrate that well-being assessment requires distinct metrics for different outcomes. Perception-based measures effectively track life satisfaction and emotional states, but monitoring severe behavioral trends like "deaths of despair" requires structural data. This confirms that well-being frameworks must be multi-layered rather than unified.

While our study highlights the superior predictive power of data-driven indicators over theoretical frameworks, it also emphasizes the enduring value of human connection. The convergence of theory and data on "belonging" as a universal anchor suggests that if governments prioritize only one policy thread in times of crisis, it should be the protection of social ties.

Areas of Further research

Moving forward, the following step of this research would be to transition from aggregate country-level means to individual country level data. This would allow researchers to escape the ecological fallacy and increase statistical power. Furthermore, since behavioral outcomes like mortality often respond slowly to socioeconomic shifts, future studies would benefit greatly from cluster-robust and lagged specifications. Finally, extending the observation window into the post-acute years of 2023 and 2024 with active testing of residual candidates will be essential to fully separate temporary pandemic shocks from lasting shifts in human well-being.

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